Sai Jagadeesh Muralikrishnan

Robotics Software Engineer | Perception, Computer Vision & 6-DoF Pose Estimation
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Summary

Robotics software engineer building perception and motion software for production robots deployed at 15+ customer sites: camera calibration, learned 6-DoF pose estimation (RT-DETR with TensorRT, FoundationPose) and outdoor UAV detection (DARPA Triage Challenge). C++/Python, ROS/ROS 2, CUDA, Jetson.

Experience

Miso Robotics Pasadena, CA
Robotics Software Engineer Sep. 2026 – Present
Associate Robotics Software Engineer (promoted after 12 months) Sep. 2025 – Sep. 2026

- Built end-to-end a prototype learned 6-DoF pipeline: RT-DETR detects fry baskets, utensils and foreign objects; FoundationPose tracks basket pose for pick-and-place at **∼15 FPS** (RT-DETR in **TensorRT FP16**); benchmarked as more precise but slower than CenterPose.
- Built ArUco-based fryer-basket misalignment detection driving adaptive pickup behaviors, cutting failed pickups from **∼30% to ∼3% (−90%)** on 10 redeployed retrofit units and removing downtime from mis-grabs and arm collisions.
- Built a ChArUco intrinsic calibration pipeline for 4K Sony IMX678 cameras, cutting RMS reprojection error **64% (0.97 → 0.35 px)**; found principal points vary **50–65 px** between units from mounting tolerance, establishing per-camera calibration.
- Debugged mis-grab and collision edge cases at **15+ customer sites** with the motion team: reproduced failures in Gazebo (ROS Noetic), ruled out human interference, and fixed planner joint limits, validated in sim and lab.
- Designed a ROS camera health validator (depth-of-field, reprojection-error and focus checks) for deployed cameras; fixed harsh-lighting exposure failures with ambient-light auto-exposure and tuned IR preprocessing for retroreflective markers.
- Integrated brand-specific PyTorch CNN food classifiers into Flippy's stack; supported retraining with field and lab data.
- Shipped **8+ ROS Noetic behavior primitives** for Flippy's 6-DoF arm with HIL test suites; since promotion, lead technical development of motion features, building a cuRobo + Isaac Sim validation workflow.

Maryland Robotics Center, University of Maryland College Park, MD
Graduate Research Assistant, DARPA Triage Challenge Sep. 2024 – May 2025

- Built perception for UAV-based casualty detection outdoors: fine-tuned YOLOv8 on remapped VisDrone data with custom augmentations (**+35% person recall**); trained lightweight PyTorch detectors for embedded deployment (**+30% mAP** under aerial occlusion); OpenCV image registration under varying lighting (**+20% feature-match rate**).

Kick Robotics College Park, MD
Robotics Software Engineer Intern May 2024 – Aug. 2024

- Calibrated Basler ToF and RealSense D435 depth cameras for outdoor navigation (Open3D checkerboard rectification, CUDA/PyTorch refinement), improving depth-map precision **12%**.
- Integrated Jetson Nano firmware, drivers and ROS 2 nodes to sync actuators and sensors (**+30%** motion smoothness); cut QA cycle time **20%** with ROS 2 validation pipelines (PyTest, GoogleTest).

TuTr Hyperloop Chennai, India
Embedded Systems Engineer Oct. 2022 – Jul. 2023

- Built multi-threaded C++ sensor-fusion pipelines over CAN (**−25% latency**) and integrated BMS, levitation and propulsion subsystems via a centralized VCU architecture, tested in a vacuum-chamber prototype.

Technical Skills

Perception: OpenCV, RT-DETR, YOLOv8, FoundationPose, CenterPose, camera calibration (ChArUco), ArUco, Open3D
ML & GPU: PyTorch, CNN classification, TensorRT, CUDA, ONNX, NVIDIA NGC
Robotics: motion planning, sensor fusion, ROS/ROS 2, Isaac Sim, cuRobo, MoveIt, Nav2, Gazebo, HIL, CAN
Sensors: Sony IMX678, RealSense D435, Basler ToF, IR
Languages & Tools: C++, Python, Bash, MATLAB, Linux, Docker, Git, GitHub Actions, PyTest, GoogleTest

Education

University of Maryland, College Park, MD – M.Eng. in Robotics, GPA 3.8/4.0 Aug. 2023 – May 2025
Grader, ENPM690 Robot Learning (imitation learning, behavioral cloning); Teaching Assistant, MSML642 Robotics
Rajalakshmi Engineering College, Chennai, India – B.E. in Mechatronics, GPA 8.7/10.0 Aug. 2018 – Jul. 2022

Awards & Publications

Maryland Robotics Center Pathways Scholarships (2024); ICDSMLA 2021 paper, Springer Nature (Best Paper, 3rd Place)